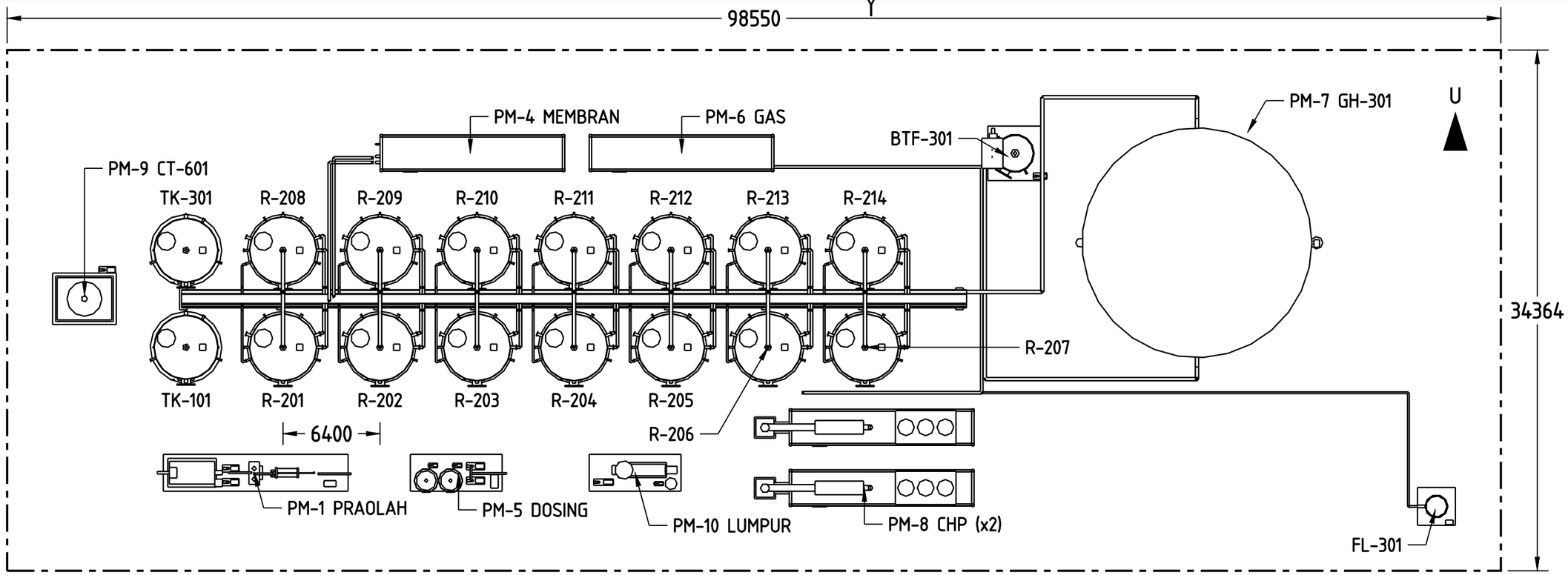
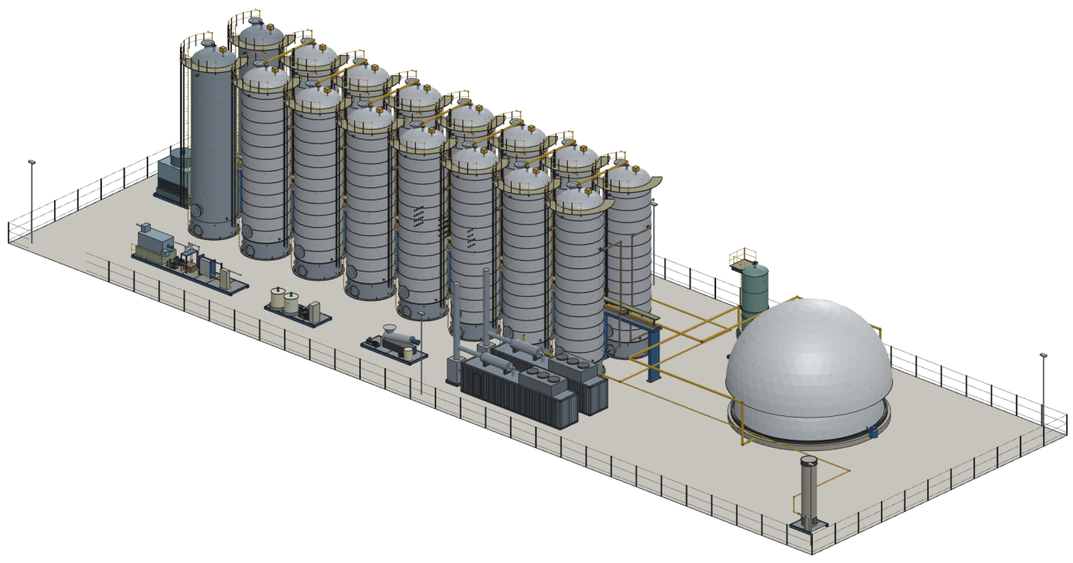


DENAH TATA LETAK (DARI MODEL 3D)

SKALA 1:300



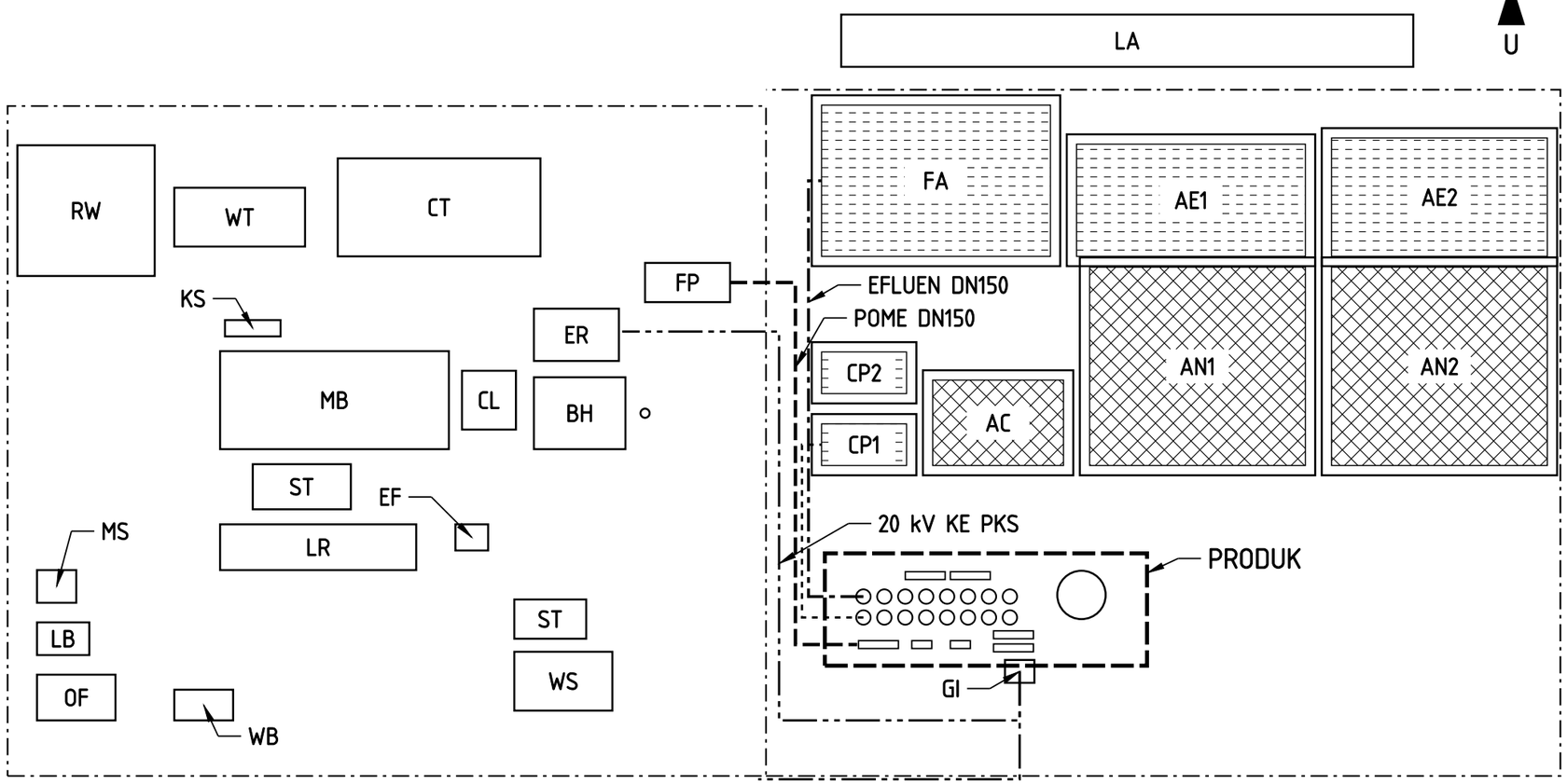
ISOMETRI SUSUNAN PABRIK (TANPA SKALA)

SET MODUL PORTABEL				
Modul	Nama	Jml	Pengiriman	Massa t
PM-1	Skid praolah POME	1	kontainer 40 ft (skid)	12.0
PM-2	Tangki penyangga umpan	1	bejana di trailer low-bed	19.0
PM-3	Reaktor AnMBR termofilik	14	bejana di trailer low-bed	266.0
PM-4	Kontainer membran + CIP	1	kontainer 40 ft HC	16.0
PM-5	Skid dosing dan pompa umpan	1	skid 20 ft	8.0
PM-6	Skid pengolahan biogas	1	kontainer 40 ft + bejana BTF	14.0
PM-7	Penampung gas + flare	1	kontainer 20 ft + skid flare	7.0
PM-8	Kontainer CHP	2	kontainer 40 ft HC	48.0
PM-9	Skid pemulih panas + menara pendingin	1	skid 20 ft + sel CT modular	9.0
PM-10	Tangki permeat + skid lumpur	1	bejana + skid 20 ft	27.0

Total 28 muatan truk, ± 426 t. Plot 98.5 x 34.4 m = 0.34 ha (termasuk jalan 3 m).

SKALA SET (DESAIN BEJANA SAMA)					
PKS	Reaktor	Genset	Neto kWe	GWh/thn	Membran m²
30 t/jam	10	2 x 600 kWe	627	4.5	634
45 t/jam	14	2 x 800 kWe	961	6.9	996
60 t/jam	19	3 x 600 kWe	1311	9.4	1314

University: UNIVERSITAS	Checked by : -	Design Lab.	Document type: General Arrangement			Rev: 0
	Author : NAMA TIM	Tata Letak Umum Set Portabel PKS 45 t TBS/jam	Unit: mm	Scale: 1 : 300	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : -		BG-GA-001			Sheet number: 1 / 9



INTERKONEKSI		
Rute (model 3D)	m	USD
POME DN150 fat pit -> PM-1	150	20955
Efluen DN150 -> kolam polishing	148	10363
Bypass darurat -> kolam pendingin	78	5437
20 kV gardu -> jaringan PLN	113	5085
20 kV ke rumah mesin PKS + trafo	256	28224
Total (masuk CAPEX)		70064

LAHAN	
Luas lahan	m²
Kolam POME sebelum	22361
Kolam tetap (polishing)	11281
Kolam dibebaskan	11081
Plot produk	3387

PEMERIKSAAN JARAK AMAN			
Jarak aman	m	min	OK
Flare -> tepi GH	16.1	15	Ya
Flare -> kontainer CHP	30.5	15	Ya
GH -> rumah boiler PKS	141.9	30	Ya
GH -> tangki CPO	203.1	30	Ya
Flare -> batas pagar lokasi	38.0	10	Ya
Reaktor -> kolam pendingin (tanah)	23.9	20	Ya
Cerobong boiler -> GH	136.0	30	Ya

CATATAN :
1. Arsir silang = kolam dibebaskan; arsir putus = air.
2. Efluen produk tetap ke kolam polishing sebelum aplikasi lahan.
3. Garis rantai = kabel 20 kV (PKS & PLN).

JALAN UMUM

250 m

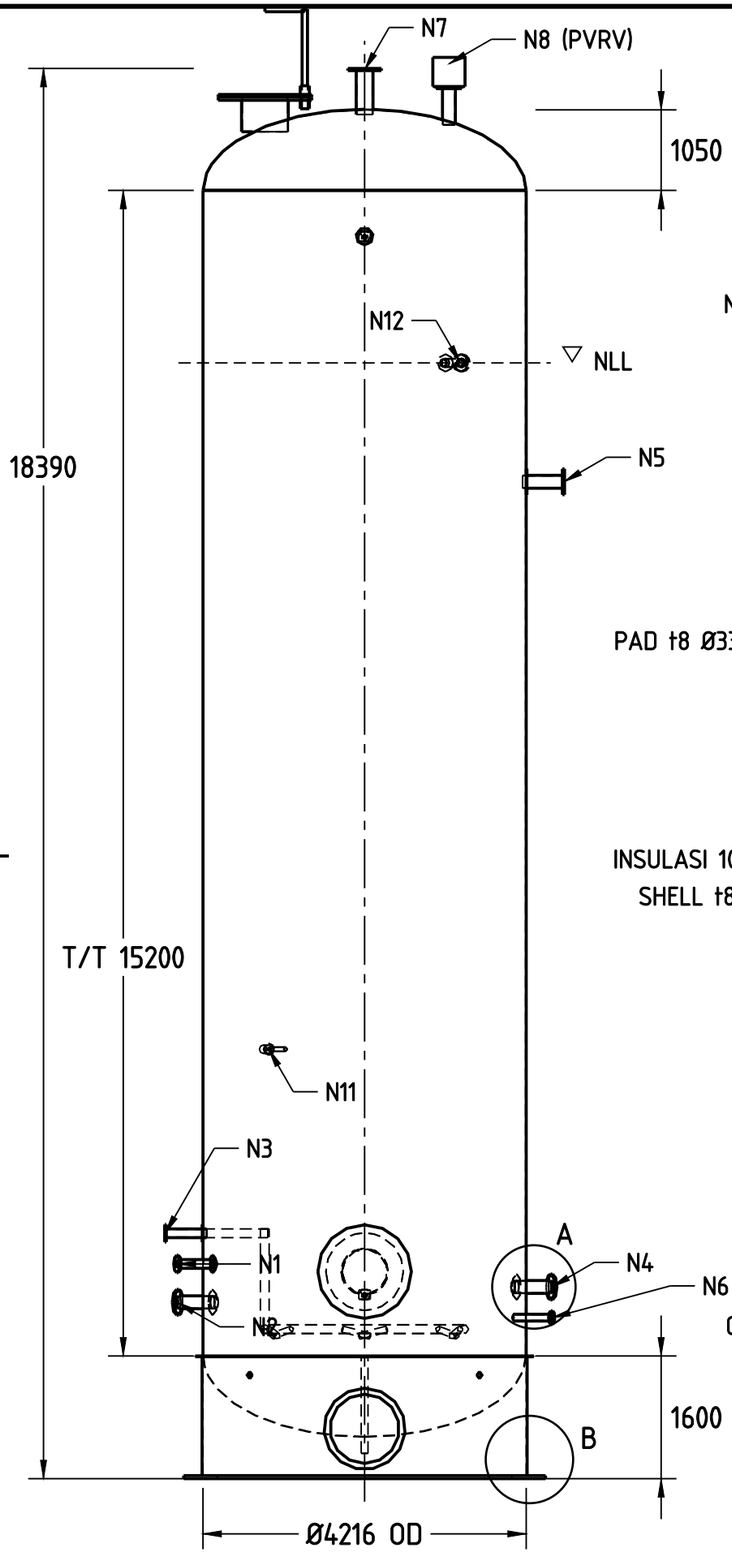
DENAH LOKASI (DARI MODEL 3D)

SKALA 1:1750

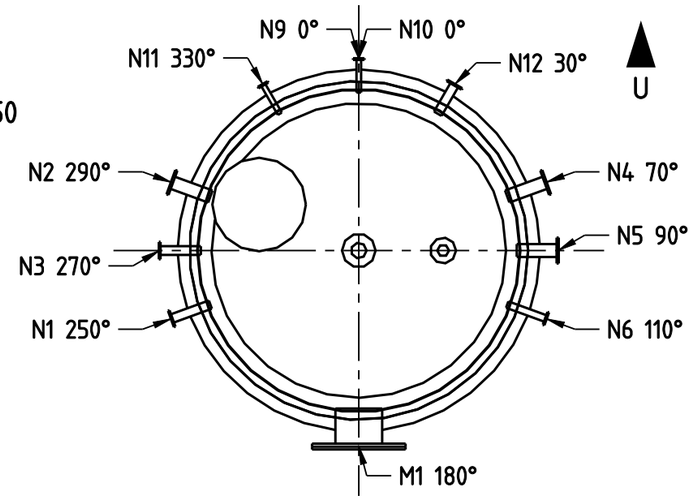
LEGENDA	
Kode	Uraian
WB	Jembatan timbang 2 x 60 t
OF	Kantor
LB	Laboratorium
WS	Bengkel
ST	Gudang
MS	Musala
LR	Loading ramp TBS 20 pintu
ST	Rebusan 3 x 30 m
MB	Pabrik utama: penebah, press, klarifikasi, k
CL	Stasiun klarifikasi (CST, tangki minyak)
KS	Silo kernel
BH	Rumah boiler 2 x 30 t uap/jam
ER	Rumah mesin: turbin uap 1,2 MW + genset dies
RW	K. air baku
WT	Pengolahan air + menara air

Kode	Uraian
CT	Tangki CPO 2 x 2.000 t
EF	Hopper TKS
FP	Fat pit (sumber POME)
CP1	K. pendingin 1 (585 m³, HRT 1.0 d)
CP2	K. pendingin 2 (585 m³, HRT 1.0 d)
AC	K. pengasaman (2,925 m³, HRT 5.0 d) (bebas)
AN1	K. anaerobik 1 (17,550 m³, HRT 30.0 d) (beba
AN2	K. anaerobik 2 (17,550 m³, HRT 30.0 d) (beba
FA	K. fakultatif (8,775 m³, HRT 15.0 d)
AE1	K. aerobik 1 (4,388 m³, HRT 7.5 d)
AE2	K. aerobik 2 (4,388 m³, HRT 7.5 d)
LA	Flatbed aplikasi lahan (efluen ke kebun)
GI	Gardu step-up 0,4/20 kV 1,25 MVA
PRODUK	Set biogas portabel, plot 3387 m²

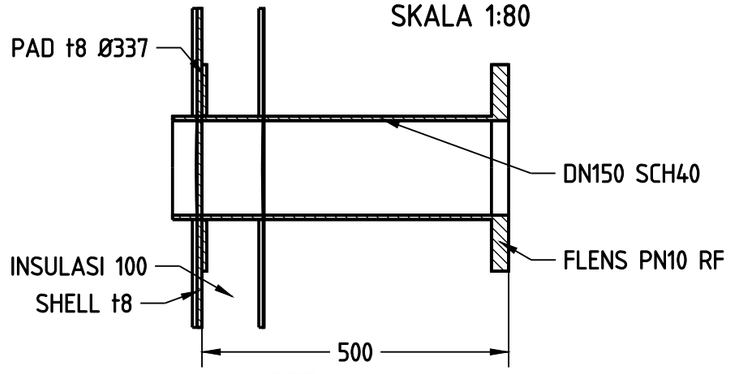
University: UNIVERSITAS	Checked by : -	Design Lab.	Document type: Site Plan			Rev: 0
	Author : NAMA TIM	Denah Lokasi PKS 45 t TBS/jam + Set Generator Biogas	Unit: mm	Scale: 1 : 1750	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : -		BG-GA-002			Sheet number: 9 / 9



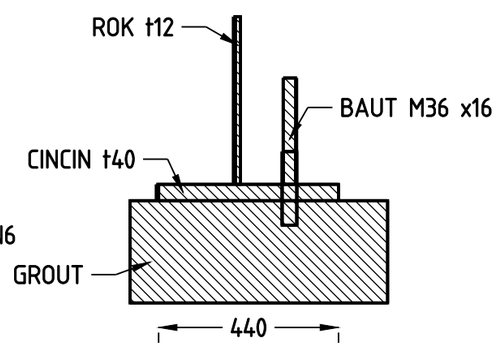
ELEVASI (DARI SELATAN)
SKALA 1:80



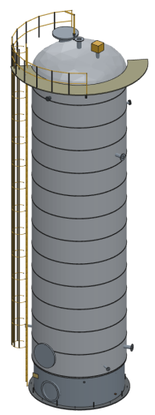
ORIENTASI NOSEL
SKALA 1:80



DETAIL A
SKALA 1:10



DETAIL B
SKALA 1:15



ISOMETRI

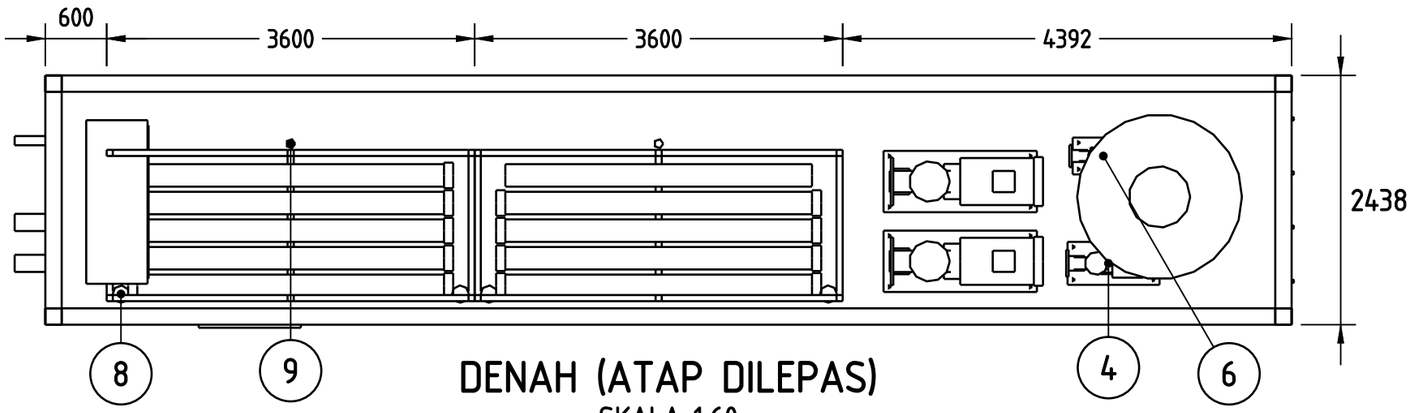
JADWAL NOSEL						
Tag	Ukuran	Rating	Layanan	Elev.	Orient.	Pad
N1	DN100	PN10 RF	Umpan reaktor (dari P-201)	2800	250°	Ø229
N2	DN150	PN10 RF	Isap jet mixing (ke P-202)	2300	290°	Ø337
N3	DN100	PN10 RF	Balik jet mixing (header nosel jet)	3200	270°	Ø229
N4	DN150	PN10 RF	Lumpur ke membran (ke P-301)	2500	70°	Ø337
N5	DN150	PN10 RF	Retentat balik (dari MB-301)	13000	90°	Ø337
N6	DN80	PN10 RF	Lumpur buangan (ke P-303)	2100	110°	–
N7	DN200	PN10 RF	Keluaran biogas	puncak	–	–
N8	DN150	PN10 RF	PVRV / relief tekanan-vakum	kepala	–	–
N9	DN50	PN10 RF	Pemancar level LT (atas / bawah)	16200	0°	–
N10	DN50	PN10 RF	Pemancar level LT bawah	2400	0°	–
N11	DN50	PN10 RF	Termometer TT / pH / NH4	5600	330°	–
N12	DN100	PN10 RF	Kaca intip muka cairan	14550	30°	Ø229
M1	DN600	PN10 RF	Lubang orang bawah	2700	180°	Ø1010
M2	DN600	PN10 RF	Lubang orang atas	kepala	–	Ø1010
N13	DN80	PN10 RF	Pengurasan (kepala bawah)	330	–	–

DATA DESAIN	
Data desain	Nilai
Fungsi / jumlah	Reaktor AnMBR termofilik, 14 unit identik + TK-101/TK-301
Kode desain / las	API 650 (acuan) + ASME VIII-1 nosel; las ASME IX
Tekanan / suhu desain	+50 / −5 mbar(g) + hidrostatik penuh; 70 °C (op. 55 °C)
Volume total / kerja	230 m³ / 194 m³, NLL 14000 mm dari BTL
Bahan shell / kepala	SA-516-70N, 8 mm (6 course x 2400), CA 3 mm
Angin / gempa	SNI 1727:2020 (40 m/s) / SNI 1726:2019, Kelas D
Insulasi / kladding	Mineral wool 100 mm 100 kg/m³ / Al 0,8 mm + sabuk
Pelapisan	Dalam: SA 2,5 + epoksi fenolik 400 µm; luar: zinc-epoksi-PU
Massa kosong / operasi	± 20.9 t / ± 219 t
Transport	rebah di low-bed, 17.3 x 4.4 m, 2 trunnion

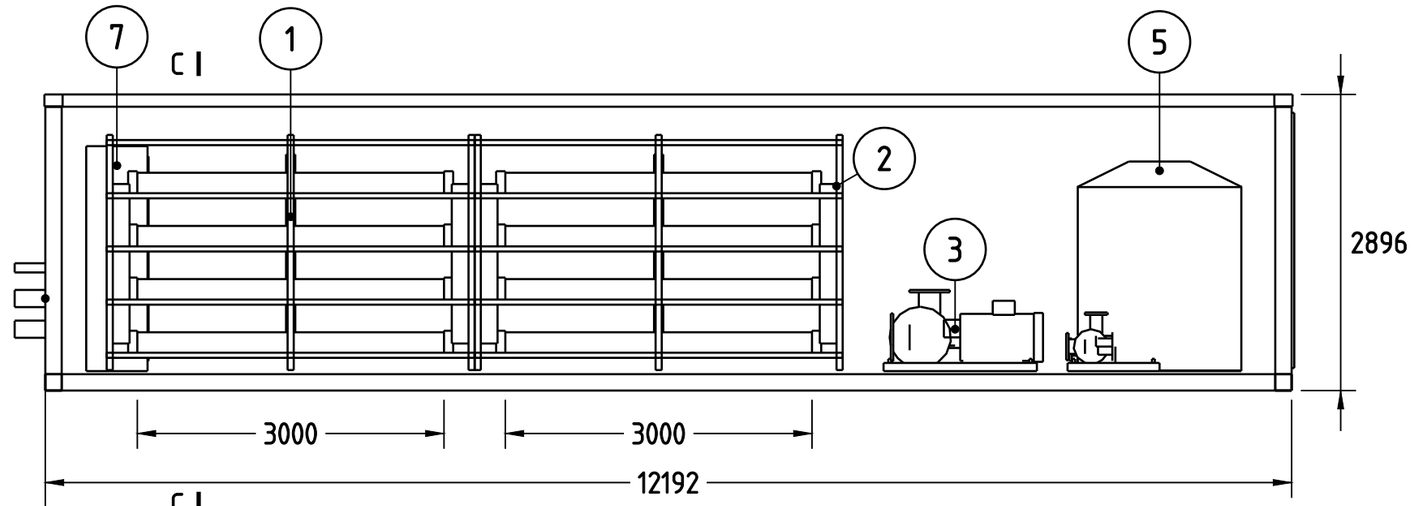
DAFTAR MATERIAL (MASSA DARI MODEL 3D)				
No	Bagian	Material	Jml	Massa kg
1	Shell (6 course)	SA-516-70N t8	1	12619
2	Kepala 2:1 atas / bawah	SA-516-70N t8	2	2311
3	Rok + rangka akses + vent	SA-516-70N t12	1	2214
4	Cincin dasar + kursi	SA-36 t40	1	1872
5	Baut angkur M36	F1554 Gr.36 galv.	16	23
6	Leher nosel + flens	SA-106B / SA-105	15	1656
7	Pad penguat	SA-516-70N t8	8	50
8	Header jet internal	SS316L DN100	1	936
9	Trunnion angkat	SA-106B DN200	2	155
10	Tangga + sangkar + platform	S275 galv., grating	1	2318
11	Insulasi mineral wool	100 kg/m³	22.1 m³	2212
12	Kladding + sabuk	Al 3003 t0,8	1	675
13	Gasket + baut flens	EPDM / A193 B7-2H	15 set	–

CATATAN :
1. Las butt penuh-tembus; RT spot 10 %, nosel & pad 100 % MT/PT.
2. Uji hidrostatik: isi penuh air 24 jam; uji pneumatik pad 1 bar.
3. Toleransi umum ISO 2768-mK; elevasi dari dasar cincin (EL 0).
4. Orientasi dari utara searah jarum jam; header jet = garis putus.

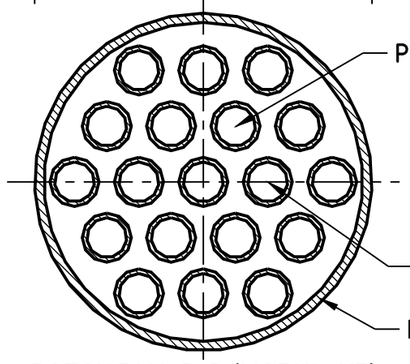
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Equipment Drawing			Rev: 0
	Author : NAMA TIM	Reaktor AnMBR Standar R-201 s/d R-214	Unit: mm	Scale: 1:80/1:10/1:15	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-V-201			Sheet number: 2 / 9



DENAH (ATAP DILEPAS)
SKALA 1:60

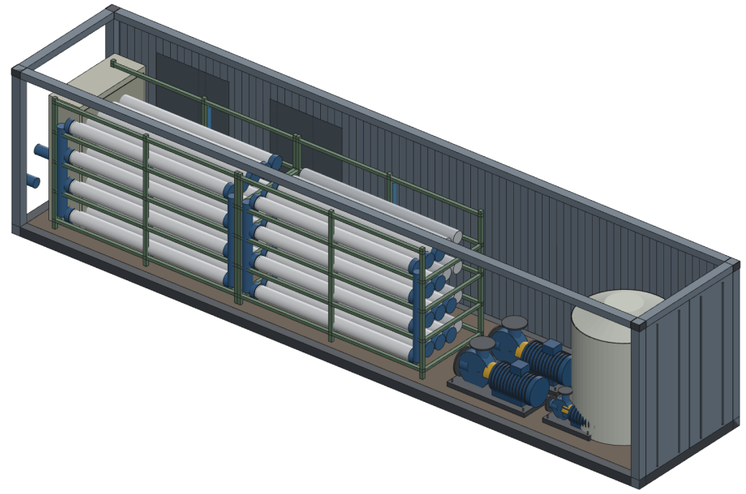


TAMPAK SAMPING (DINDING DILEPAS)
SKALA 1:60



POTONGAN C-C (MODUL UF)
SKALA 1:4

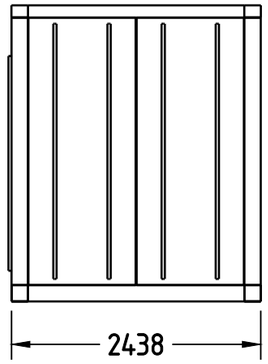
CATATAN :
1. 2 bank x 5 kolom x 4 tingkat (40 posisi), 34 terpasang + 6 cadangan.
2. Fluks 15 LMH, luas 996 m², cross-flow 4 m/s.
3. Lorong servis 850 mm di sisi pompa.
4. Dinding kiri & atap dilepas untuk kejelasan.
5. CIP: NaOCl 500 ppm + asam sitrat 2 %, tiap 30 hari.



ISOMETRI POTONG (TANPA SKALA)

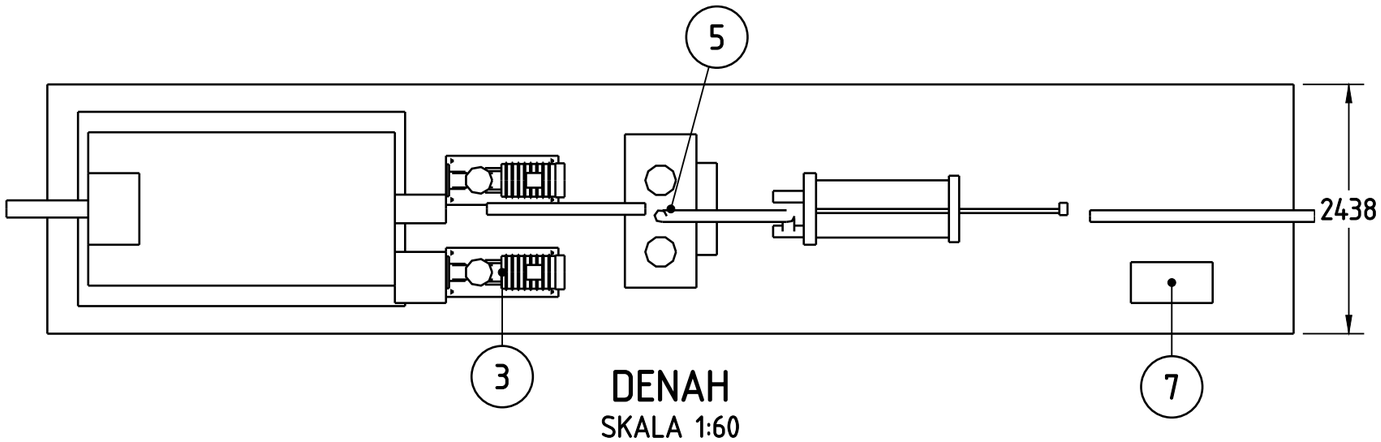
DAFTAR BAGIAN					
No	Tag	Uraian	Bahan	Jml	kg
1	MB-301	Modul UF tubular Ø220x3000, 30 m²	PVDF/PVC-U	34	2108
2	–	Rak modul 2 bank	S275 galv.	2	1450
3	P-301A/B	Pompa cross-flow 149 m³/jam	CF8M, 30 kW	2	2x520
4	P-302	Pompa permeat	CF8M, 5,5 kW	1	140
5	TK-302	Tangki CIP 4 m³	SS304	1	380
6	P-304	Pompa CIP	CF8M, 4 kW	1	110
7	–	Panel MCC + PLC	IP54, VSD	1	650
8	–	Header umpan/retentat DN150	SS316L Sch10S	4	4x60
9	–	Header permeat DN80	PVC-U PN10	2	2x15
10	–	Nosel dinding DN150/80	SS316L	3	3x25
11	–	Kontainer ISO 40 ft HC	Corten, insul.	1	4200
12	–	Pipa, katup, kabel, CIP	–	1 lot	900

Total kering ± 12,1 t; transport ± 16 t (berisi air uji).

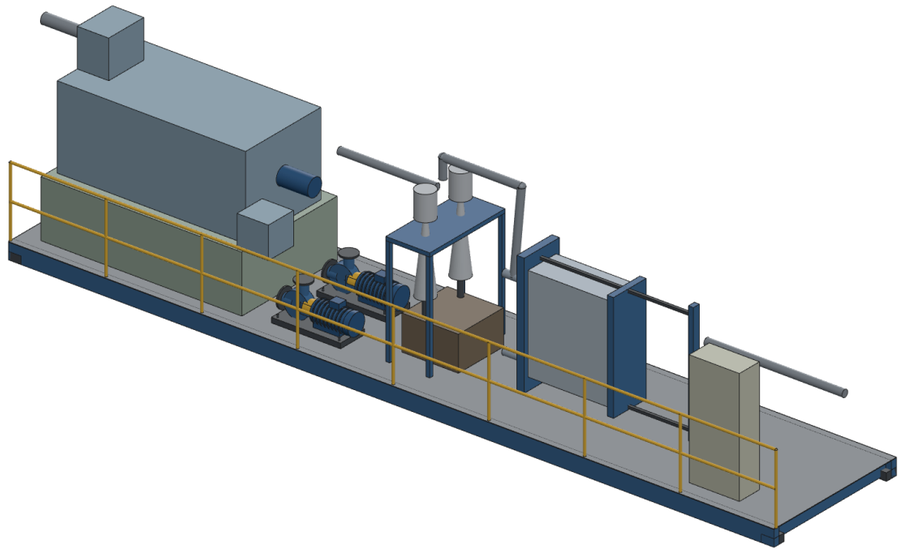


TAMPAK UJUNG
SKALA 1:60

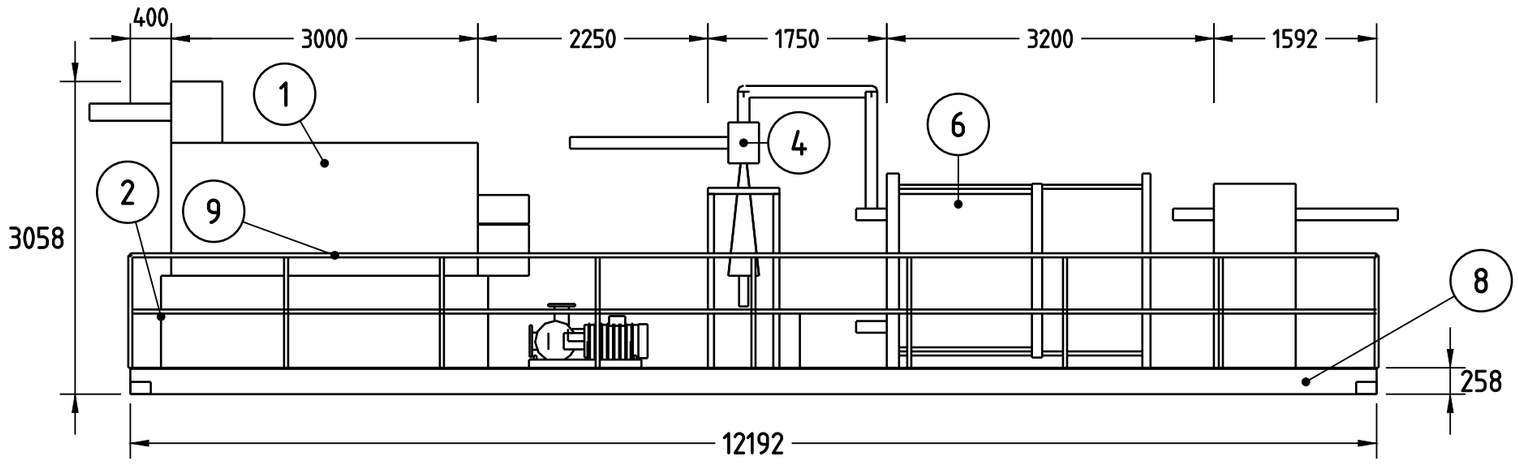
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing	Rev: 0
	Author : NAMA TIM	PM-4 Kontainer Membran UF (MB-301, P-301, P-302, TK-302)	Unit: mm / Scale: 1 : 60 / Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-004	Sheet number: 3 / 9



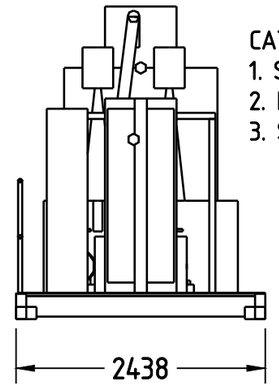
DENAH
SKALA 1:60



ISOMETRI (TANPA SKALA)



TAMPAK DEPAN
SKALA 1:60



TAMPAK UJUNG
SKALA 1:60

CATATAN :
1. Seluruh alat dirakit & diuji di bengkel.
2. Pengiriman: 1 truk trailer 40 ft, ± 12 t.
3. Sambungan: POME masuk DN150 (kiri), POME keluar DN100 ke TK-101 (kanan).

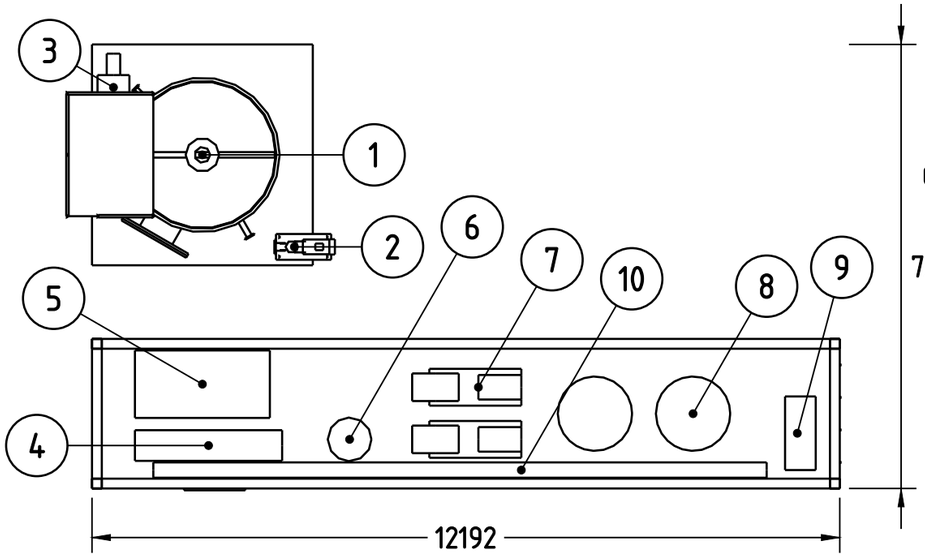
DAFTAR BAGIAN

No	Tag	Uraian	Bahan	Jml	kg
1	SC-101	Saringan drum Ø900x2600, 0,75 mm	SS304	1	1450
2	–	Bak sump 3 m³	SS304 t4	1	620
3	P-101A/B	Pompa desander H 20 m, VSD	CD4MCu, 11 kW	2	2x310
4	HC-101	Hidrosiklon Ø300	PU-lined CS	2	2x140
5	–	Bak pasir 0,4 m³	S275 galv.	1	180
6	E-101	PHE 812 kWth	SS316/EPDM	1	1850
7	–	Panel kontrol lokal	IP65	1	220
8	–	Skid UNP 250 12192x2438	S275, cat	1	3900
9	–	Pegangan tangan Ø42,4	S275 galv.	1	240
10	–	Pipa, katup, kabel	SS304 / PVC	1 lot	1100

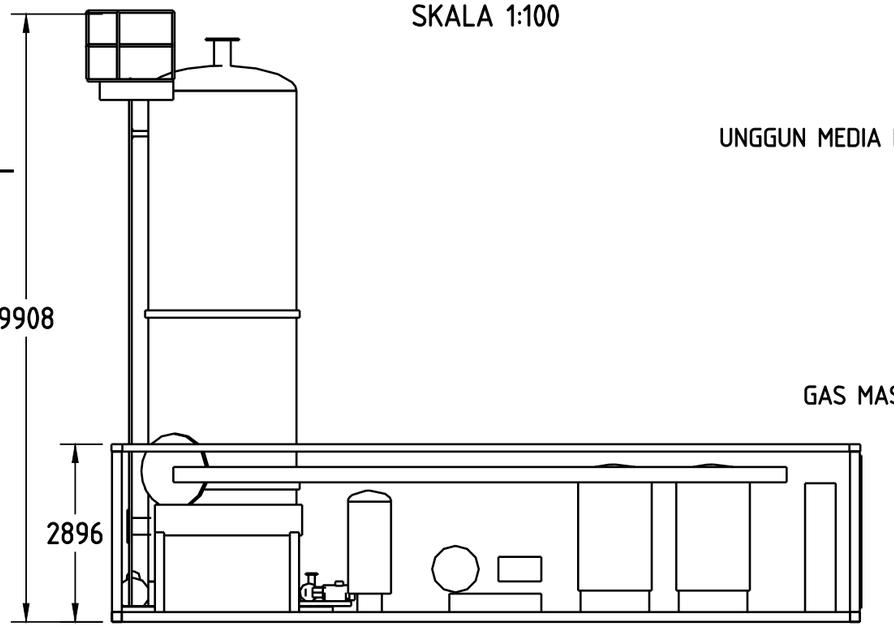
SAMBUNGAN SKID

Tag	Ukuran	Layanan	Dari → ke
N-A	DN150	POME panas masuk (85 °C)	PKS → SC-101
N-B	DN100	POME tersaring ke TK-101	E-101 → TK-101
N-C	DN80	Air pendingin masuk / keluar	CT-601 ↔ E-101
N-D	DN50	Underflow pasir	HC-101 → bak
N-E	–	Daya 400 V, 3 fasa, 45 kVA	MCC → panel

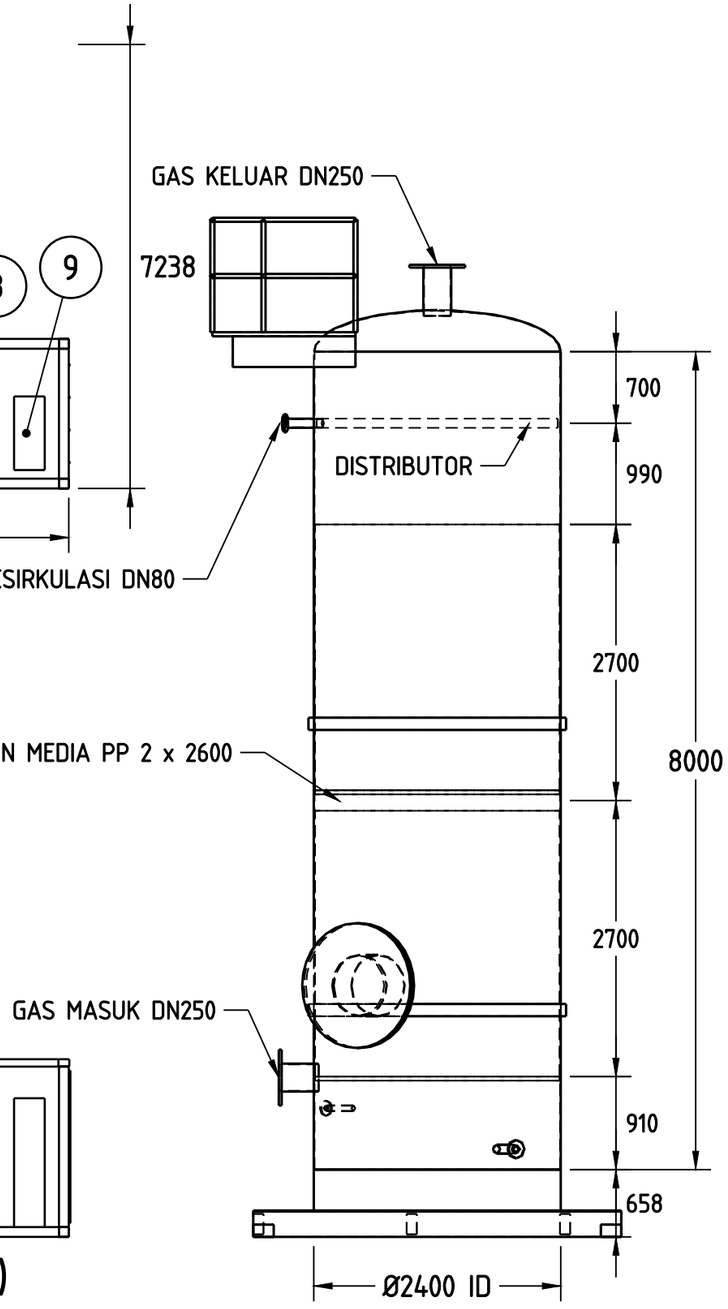
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing	Rev: 0
	Author : NAMA TIM	PM-1 Skid Praolah POME (SC-101, HC-101, E-101)	Unit: mm / Scale: 1 : 60 / Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-001	Sheet number: 4 / 9



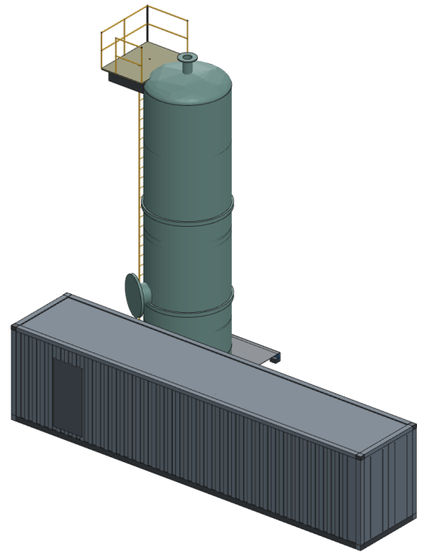
DENAH (ATAP DILEPAS)
SKALA 1:100



TAMPAK DEPAN (DINDING DILEPAS)
SKALA 1:100



BTF-301 (ELEVASI)
SKALA 1:60



ISOMETRI (TANPA SKALA)

DAFTAR BAGIAN

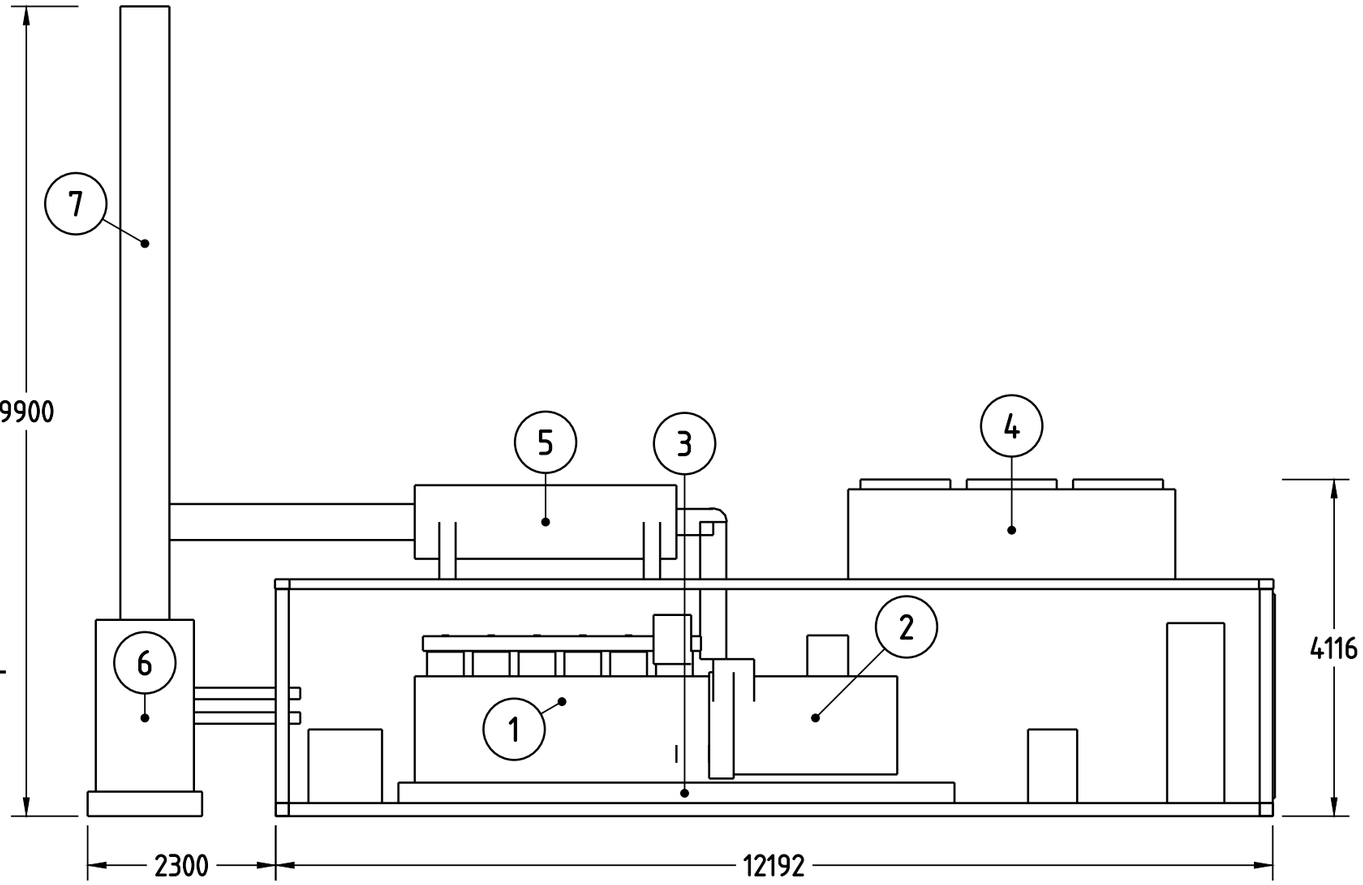
No	Tag	Uraian	Bahan	Jml	kg
1	BTF-301	Bio-trickling filter Ø2400x8000	FRP vinilester	1	3800
2	P-305	Pompa resirkulasi nutrien	PP, 1,5 kW	1	60
3	K-301	Blower udara BTF	Al, 0,75 kW	1	45
4	HX-301	Pendingin gas shell-tube	SS316L	1	650
5	CH-301	Chiller paket	R-32, 15 kWe	1	780
6	V-301	KO pot Ø700 + demister	SS304	1	210
7	K-302A/B	Blower biogas 537 m³/jam	Al, ATEX 18,5 kW	2	2x420
8	AC-301A/B	Filter karbon aktif Ø1200	CS epoksi + KI-AC	2	2x1650
9	–	Panel kontrol + analis gas	IP54, ATEX	1	380
10	–	Header gas DN250	SS304	1	420
11	–	Kontainer 40 ft HC, ATEX z2	Corten	1	4300
12	–	Skid BTF 3600 + tangga	S275 galv.	1	1650

NOSEL BTF-301

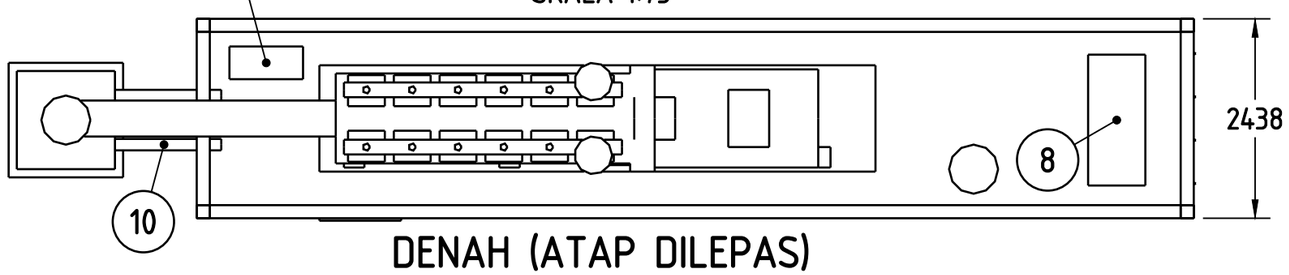
Tag	Ukuran	Layanan BTF-301
G1	DN250	Gas masuk (bawah)
G2	DN250	Gas keluar (puncak)
A1	DN50	Udara dari K-301
L1	DN80	Resirkulasi ke distributor
L2	DN80	Sump / blowdown
MH	DN600	Lubang orang

CATATAN :
1. BTF dikirim terpisah di trailer.
2. H_2S 2887 → 34 ppm.
3. Kontainer: ventilasi paksa + detektor gas.

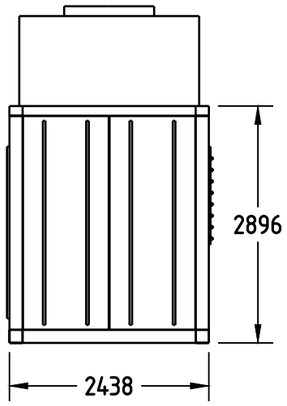
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing		Rev: 0
	Author : NAMA TIM	PM-6 Skid Pengolahan Biogas (BTF-301, K-302, AC-301)	Unit: mm	Scale: 1:100 / 1:60	Creation date: 15/10/2026 / Size: A3
	Eng. Approved by : –		BG-M-006		Sheet number: 5 / 9



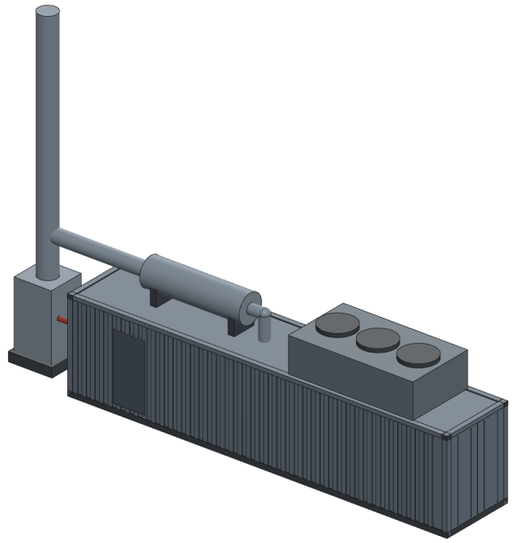
TAMPAK DEPAN (DINDING DILEPAS)
SKALA 1:75



DENAH (ATAP DILEPAS)
SKALA 1:75



TAMPAK UJUNG
SKALA 1:75



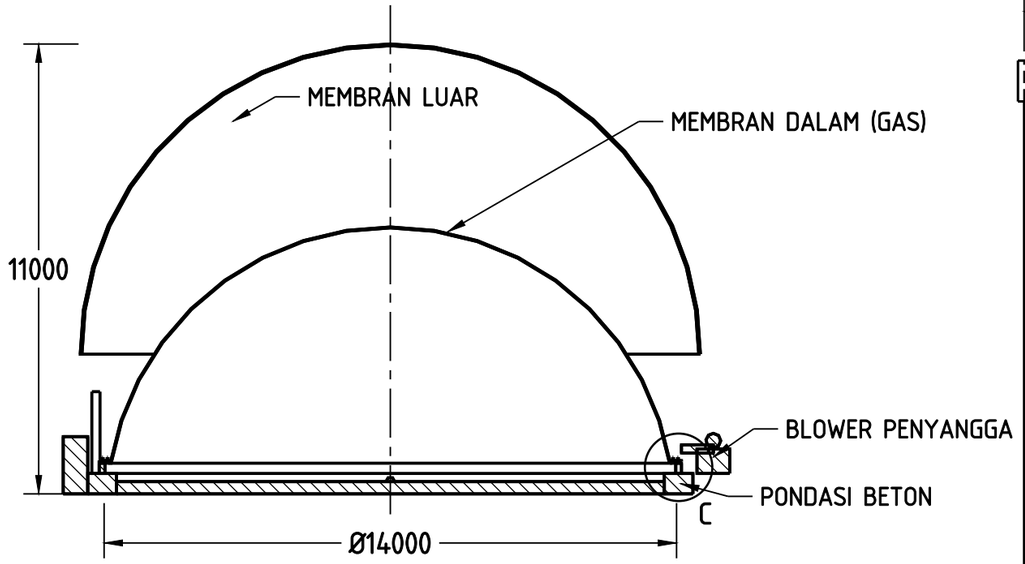
ISOMETRI (TANPA SKALA)

DAFTAR BAGIAN (PER UNIT)					
No	Tag	Uraian	Bahan	Jml	kg
1	GE-401	Mesin gas V-12, 800 kWe	blok besi cor	1	7800
2	–	Generator sinkron 400 V 50 Hz	IP23, H	1	3100
3	–	Rangka dasar + peredam	S355 + karet	1	1400
4	–	Radiator / dump cooler	Al/Cu, 3 kipas	1	1250
5	–	Silencer Ø900x3200	CS, 35 dB(A)	1	650
6	HX-401	Pemulih panas gas buang	SS316L/CS	1	1900
7	ST-401	Cerobong Ø600 x 9,9 m	CS galv.	1	850
8	–	Panel kontrol + sinkron	IP54, 1600 A	1	900
9	–	Gas train + regulator	DN100, ATEX	1	260
10	–	Pipa HWS/HWR DN80	CS Sch40 insul.	2	2x60
11	–	Kontainer 40 ft HC, 75 dB(A)	Corten + insul.	1	5200

DATA KINERJA GE-401	
Data genset	Nilai
Daya listrik terpasang / operasi	800 kWe / 544 kWe (beban 71 %)
Efisiensi listrik (beban parsial)	38.0 %
Bahan bakar per unit	1431 kW (LHV)
Panas pulih (jaket + gas buang)	720 kWth
Suhu gas buang masuk HX / cerobong	476 °C / 150 °C
Emisi (Permen LHK 11/2021)	NOx ≤ 500, CO ≤ 1000 mg/Nm³

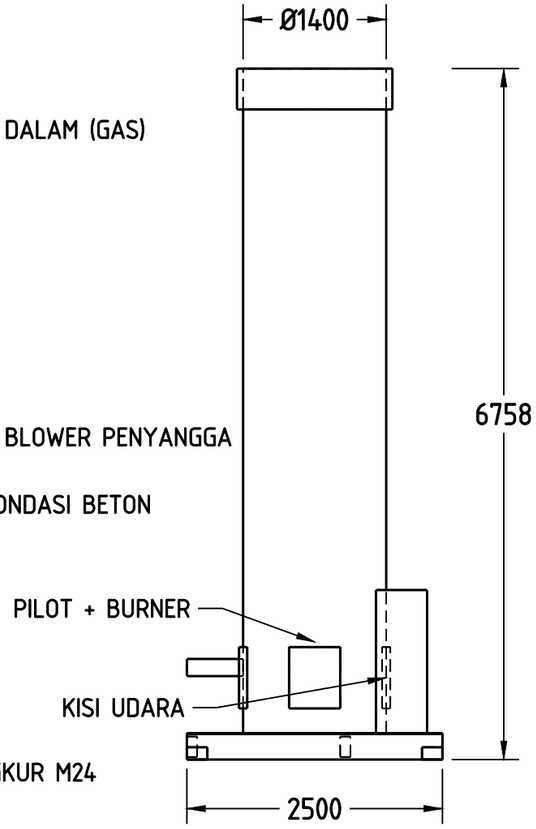
CATATAN :
1. Dua unit identik GE-401 / GE-402 (2 kontainer).
2. Bahan bakar: biogas H_2S ≤ 50 ppm dari PM-6.
3. HX-401 & ST-401 dikirim sebagai skid PM-9.
4. Massa angkut ± 24 t per kontainer.

University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing	Rev: 0
	Author : NAMA TIM	PM-8 Kontainer CHP 800 kWe + HX-401 / ST-401 (PM-9)	Unit: mm / Scale: 1 : 75 / Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-008	Sheet number: 6 / 9



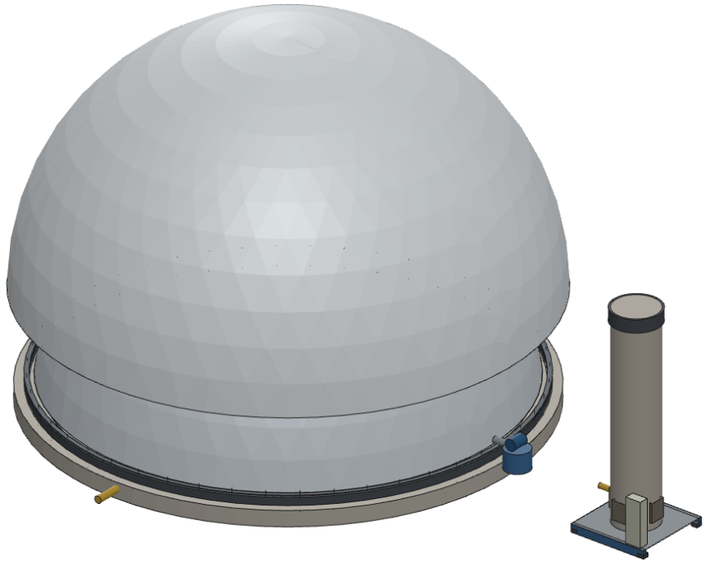
POTONGAN A-A GH-301

SKALA 1:150



FL-301 (ELEVASI)

SKALA 1:60

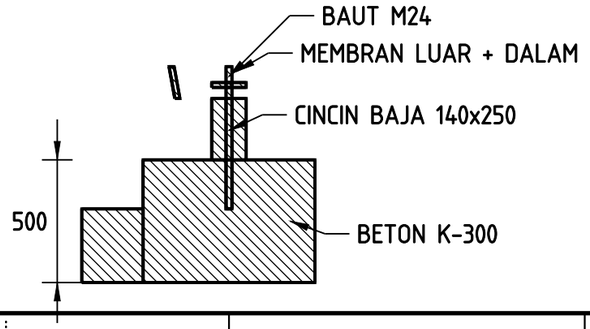


ISOMETRI (TANPA SKALA)

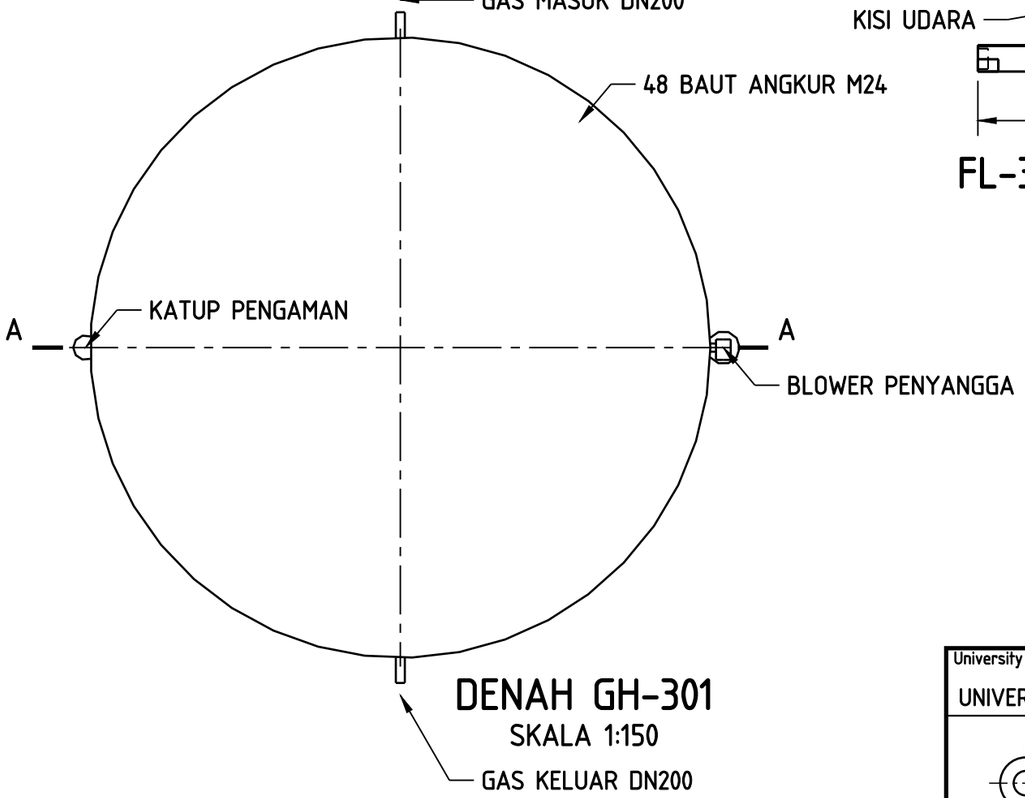
DATA DESAIN		
Tag	Parameter	Nilai
GH-301	Volume gas berguna	1135 m³ (± 2 jam P95)
	Volume kubah total	1414 m³, Ø14.0 x 10.5 m
	Membran luar / dalam	PVC-poliester 1100 g/m² / PE kedap metana
	Tekanan operasi	2 – 3 mbar(g), blower udara penyangga
	Pondasi	Cincin beton Ø14800, lantai beton 300 mm
	Pengiriman	membran terlipat di 1 kontainer 20 ft
FL-301	Kapasitas	624 Nm³/jam, suhu nyala ≥ 1000 °C
	Tipe	Flare tertutup, pilot otomatis, skid 2,5 x 2,5 m

CATATAN :

1. Membran luar dijaga tegang oleh blower udara penyangga.
2. Membran dalam naik-turun sesuai isi gas (LIC-401).
3. Flare menyala otomatis saat GE berhenti / GH penuh (PIC-407).
4. Jarak aman GH ke flare ≥ 15 m (lihat tata letak BG-GA-001).



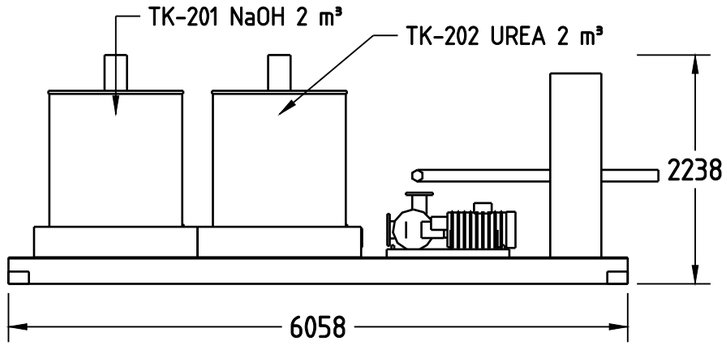
DETAIL C
SKALA 1:25



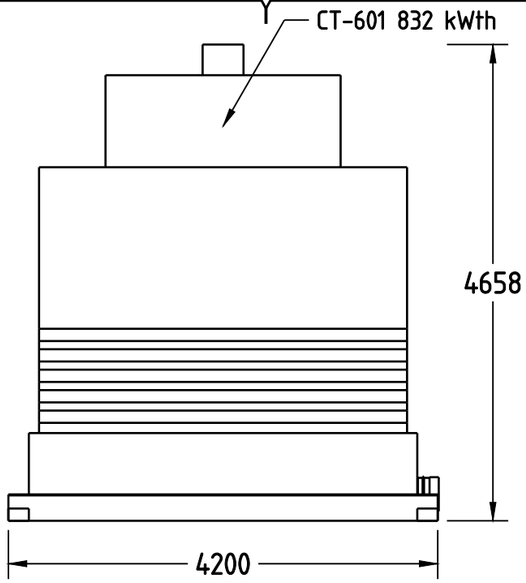
DENAH GH-301

SKALA 1:150

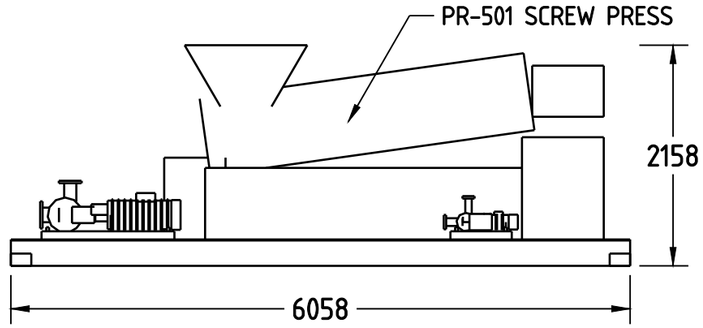
University: UNIVERSITAS	Checked by : -	Design Lab.	Document type: Assembly Drawing		Rev: 0
	Author : NAMA TIM	PM-7 Penampung Gas GH-301 + Flare Tertutup FL-301	Unit: mm	Scale: 1:150 / 1:60	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : -		BG-M-007		Sheet number: 7 / 9



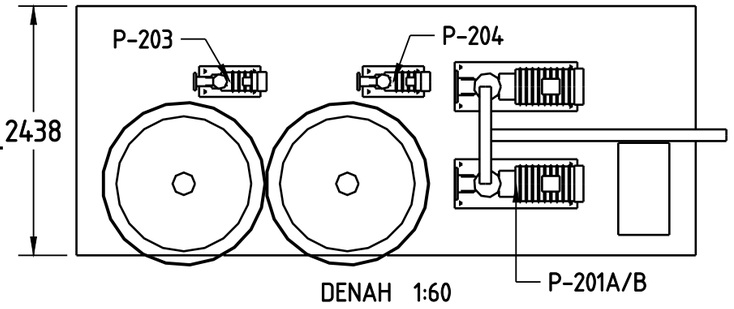
PM-5 TAMPAK DEPAN 1:60



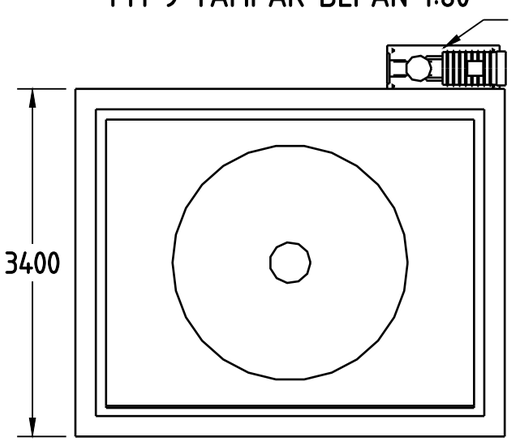
PM-9 TAMPAK DEPAN 1:60



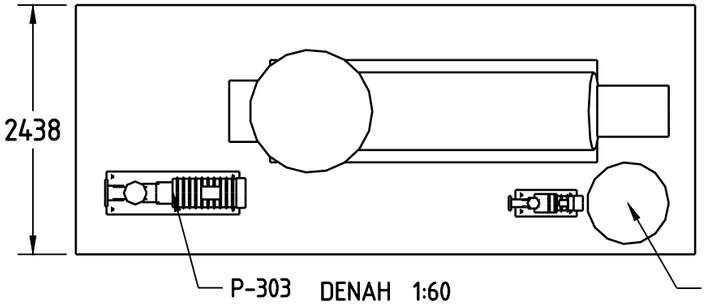
PM-10 TAMPAK DEPAN 1:60



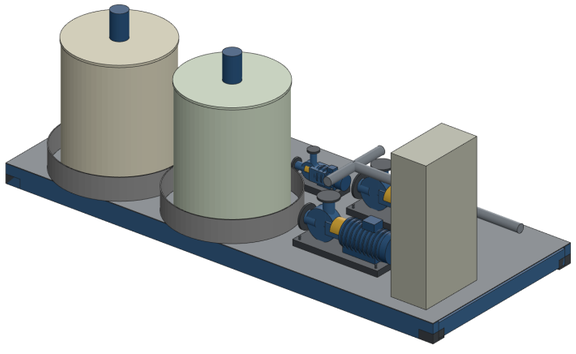
DENAH 1:60



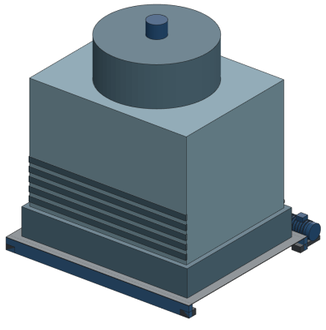
DENAH 1:60



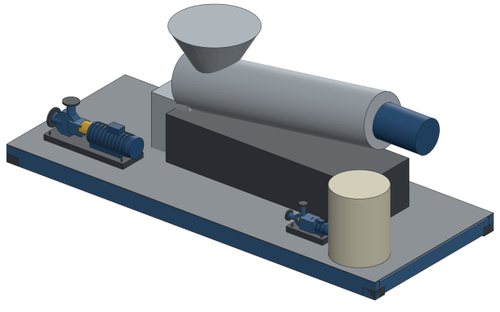
DENAH 1:60



ISOMETRI PM-5



ISOMETRI PM-9



ISOMETRI PM-10

DAFTAR BAGIAN SKID				
Skid	Tag	Uraian	Bahan / daya	kg
PM-5	TK-201/202	Tangki NaOH / urea 2 m³	HDPE + bund	2x260
	P-201A/B	Pompa umpan reaktor	CF8M, 7,5 kW	2x260
	P-203/204	Pompa dosing	PVDF, 0,37 kW	2x45
	-	Panel dosing + skid 20 ft	IP65 / S275	2080
PM-9	CT-601	Menara pendingin 832 kWth	FRP, 7,5 kW	2900
	P-601	Pompa air pendingin	GG25, 11 kW	210
	-	Skid 4200 x 3400	S275 galv.	1500
PM-10	P-303	Pompa lumpur (progressive cavity)	CS/NBR, 4 kW	380
	PR-501	Screw press lumpur	SS304, 5,5 kW	2600
	TK/P-polimer	Tangki + pompa polimer 1 m³	HDPE / PVDF	160
	-	Skid 20 ft + chute kue	S275 galv.	1900

University: UNIVERSITAS	Checked by : -	Design Lab.	Document type: Assembly Drawing			Rev: 0
	Author : NAMA TIM	Skid PM-5, PM-9 dan PM-10 (Dosing, CT-601, Lumpur)	Unit: mm	Scale: 1 : 60	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : -		BG-M-005			Sheet number: 8 / 9